

Programming in Python for Friends and Relations—Part 14



A Musical Button

As we discovered last month, *csound* can be used to create new instruments. Now, how about creating what sound like 'real' instruments?

*S*oundFont is a trademark of E-mu. In the 1990s, E-mu came up with the idea of storing samples of the sounds of real instruments and replaying them with the desired effects. The aim was that SoundBlaster cards from Creative Labs should be able to play MIDI files realistically.

A composer can compose music and explore it with an 'orchestra' of instruments even if he is alone in his studio. We, however, are after a more limited goal.

The TamTam activities included in the OLPC/Sugar environment are really great tools for children to explore music and sounds. We will try to explore and understand the underlying relationship

between *csound* and SoundFonts.

On Fedora 10, SoundFonts are available in `/usr/share/soundfonts`. PCLite.sf2 is a part of the package, PersonalCopy-Lite-soundfont. Incidentally, this SoundFont is also required by Timidity++, a software player of MIDI files.

On Ubuntu 8.10, the SoundFonts, `/usr/share/sounds/sf2/FluidR3_GM.sf2`, are installed by the *fluid-soundfont* packages.

The CSD file

Create the following *sf2.csd* file, with the definition of an instrument that plays SoundFont notes:

```
<CsoundSynthesizer>
<CsOptions>
```